Learning a Decoder-Free Depth Encoder for Fixed-Focus 3D Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Computational 3D fluorescence microscopy relies on a reconstruction prior wherever the measurements carry little information, and a strong prior can return a sharp volume whose depth the measurements do not support. We learn an optical encoder that records depth, and evaluate it with several decoders, none of which is used in its training. A plane wave is scanned in angle through a fixed phase plate placed before an intermediate image plane, so that each depth receives its own sequence of excitation patterns while the specimen and the detection focus stay still. We train the plate with a specimen-free objective that decorrelates the detector-integrated measurement signatures of different depths, with no reconstruction in the loop. In simulations of mouse pyramidal neuron volumes at eight-fold measurement compression and equal excitation dose, the learned plate lowers the depth error of total variation and of pretrained and neuron-trained plug-and-play reconstruction by 30–54 % relative to a random plate on 60 volumes, with cell-level intervals that exclude zero and a lower error on 179 of 180 volume–decoder pairs; a diffusion-based decoder shows the same direction. Its objective gives a 12–14 % lower depth error than coherence and Fisher-information objectives trained on the same budget, and the plate narrows conformal depth intervals by 27 %. An in-focus structured-illumination encoder leaves every tested decoder at the error of a depth-blind estimate.

1. Introduction

Fixed-focus three-dimensional fluorescence microscopy encodes depth into planar camera images and recovers the volume computationally. The decoders that perform this recovery are changing quickly, from hand-crafted regularisers to plug-and-play denoisers and diffusion models [8, 30, 37, 38]. Where the measurements carry little depth information, a stronger prior still returns a plausible depth, but the data do not support it [4, 10]. In neuronal imaging depth is the quantity that matters: whether a dendrite lies above or below its neighbour decides which structures it can touch.

The depth information therefore has to be recorded by the optics. We study a fixed-focus microscope that encodes depth in the excitation (Fig. 2). A plane wave is scanned in angle through one fixed phase plate placed a short distance before an intermediate image plane; a high-NA relay maps the field into the specimen, and a low-NA camera at fixed focus records one image per angle. The camera keeps a wide field, but its depth of field at NA 0.1, about $50\,\mu\mathrm{m}$, gives it weak axial discrimination across the $10\,\mu\mathrm{m}$ volume. A tilt translates the diffracted pattern of the plate by an amount that grows with propagation distance, so each depth sees its own sequence of excitation values without axial scanning or refocusing.

The plate is the learnable element. Learned optics are usually optimised end-to-end with a reconstruction network [12, 23, 29], which matches the hardware to that network, although the hardware is fixed once fabricated and decoders

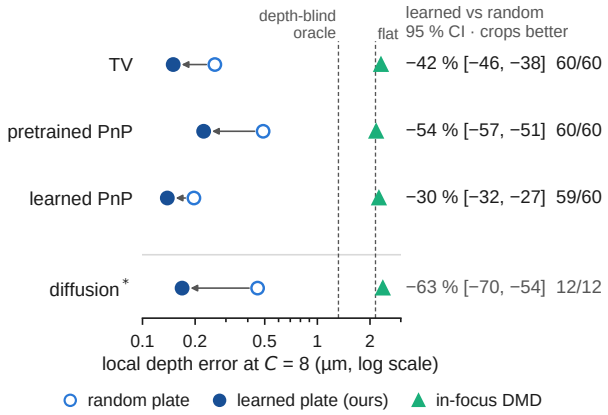


Figure 1. **One learned plate, four decoders.** Mean local depth error of 60 mouse pyramidal neuron volumes from six cells at eight-fold compression and equal excitation dose; each decoder reconstructs the same measurements of each encoder. The plate is trained without a decoder and lowers the depth error of every tested decoder (right: change against the random plate it started from, cell-bootstrap 95 % interval, crops improved). *Diffusion (12 volumes) missed our prespecified criterion for a competitive decoder on the random plate, so its comparison is descriptive. The in-focus DMD stays near the error of a flat estimate with every decoder (DMD and reference lines: the 12 prespecified volumes).

keep changing. Decoder-free criteria, such as Fisher information, coherence or estimated mutual information [25, 28, 33], avoid this coupling. We train the plate with a decoder-free objective of this kind, designed for the encoder: it decorrelates the detector-integrated signatures that a voxel leaves across scan angles at different depths, computed through a voxel-integrated vectorial model of the optics, with no specimen and no reconstruction in the loop. We then ask whether the learned plate improves depth recovery for decoders of different kinds, and whether the improvement survives a calibrated, per-volume error statement.

On simulated measurements of neuron volumes, the learned plate lowers the depth error of total variation and of two plug-and-play decoders by 30–54 % on 60 test volumes (Fig. 1), and a diffusion-based decoder shows the same direction. It narrows conformal depth intervals by 27 %, outperforms coherence and Fisher-information objectives trained on the same budget, and keeps its gain under calibrated fabrication and alignment errors. An in-focus structured-illumination encoder, evaluated identically, leaves every decoder at the error of a depth-blind estimate. Our contributions are:

1. **A fixed-focus depth encoder.** Angle scanning through a defocused phase plate gives every depth its own excitation sequence; a voxel-integrated vectorial operator models it and is assembled efficiently.
2. **A decoder-free training objective for this encoder.** Squared correlations between the depth signatures of each lateral position, which need no specimen. Under the same start and budget it lowers the depth error by 12–14 % relative to plates trained for worst-case coherence or Fisher information.
3. **Evaluation across decoders and with calibrated intervals.** At equal dose, the learned plate improves depth under four decoders, replicates with a second training seed, narrows conformal depth intervals from ± 0.65 to $\pm 0.47\,\mu$m, and tolerates calibrated fabrication and alignment errors.

2. Related work

Depth encoding in fluorescence microscopy. Light-field microscopy, DiffuserCam and Miniscope3D encode volumetric information in the detection response [3, 6, 33]. Excitation coding offers a complementary route. Yeh *et al*. [34] reconstruct three-dimensional fluorescence using translated speckle illumination and axial sample scanning, while Pascucci *et al*. [24] exploit axial speckle decorrelation to recover volumes from a two-dimensional scan. Yilmaz *et al*. [35] vary the incident angle through a fixed scattering layer for two-dimensional recovery, and Jiang *et al*. [14] use tilted pattern projection to produce depth-dependent lateral shifts. Our acquisition combines these mechanisms: the incident angle is scanned through a fixed, defocused phase plate, with a stationary specimen and fixed camera focus, and the plate is learned.

Learning optical encoders. End-to-end methods optimise optics jointly with a reconstruction network, for extended depth of field [29], monocular depth [7, 32], 3D localisation [23], coded illumination [17] and compressive fluorescence microscopy [12]. The learned optics are then matched to one decoder. Decoder-free design instead scores the measurements: Shechtman *et al*. [28] maximise Fisher information, Miniscope3D [33] penalises the coherence of depth-dependent point-spread functions, building on coherence minimisation in sensing design [9], and IDEAL [20, 25] optimises an estimate of mutual information. Our objective belongs to this family but is built for an excitation encoder: it decorrelates the detector-integrated signatures that a voxel leaves across scan angles at different depths, and requires no specimen examples. We compare it with coherence and Fisher-information objectives trained under the same budget, and report its effect under several decoders rather than one.

Priors, hallucination and guarantees. Plug-and-play methods insert learned denoisers into iterative solvers [15, 30, 37], and diffusion models now provide generative priors for inverse problems [8, 16, 38]. A prior can produce structure that the forward model does not determine [4, 10, 13], and posterior spread from a learned model is not a calibrated error bound [22]. Distribution-free methods calibrate reconstruction errors on held-out data [1]; conformal risk control [2] bounds the expected value of a monotone loss. We use it to calibrate depth placement, and use set-membership bounds over measurement-consistent volumes [11, 21] to examine what the data determine without the learned prior.

Forward modelling. Incoherent structured-illumination models combine excitation intensity with the emission response [34], and wave-optics models account for diffraction in volumetric imaging [6]. We integrate fine-grid optical responses over reconstruction voxels, with a vectorial high-NA relay [19, 26]. Our reconstruction uses established components, adapting multi-slice fusion [18] to three-dimensional ADMM; we do not propose a new reconstruction algorithm.

3. Angle-scanned phase-encoded 3D imaging

Acquisition. We vary the illumination direction across two angular coordinates while keeping the phase plate fixed (Fig. 2). For transverse position $\mathbf{r} = (x, y)$, the incident plane wave is $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^{\top}\mathbf{r}}$, with $\mathbf{s}_t = \cos\theta_t(\cos\alpha_t, \sin\alpha_t)$, where α_t is the azimuth and θ_t parametrises the direction in the vertical plane containing that azimuth. The corresponding forward direction is

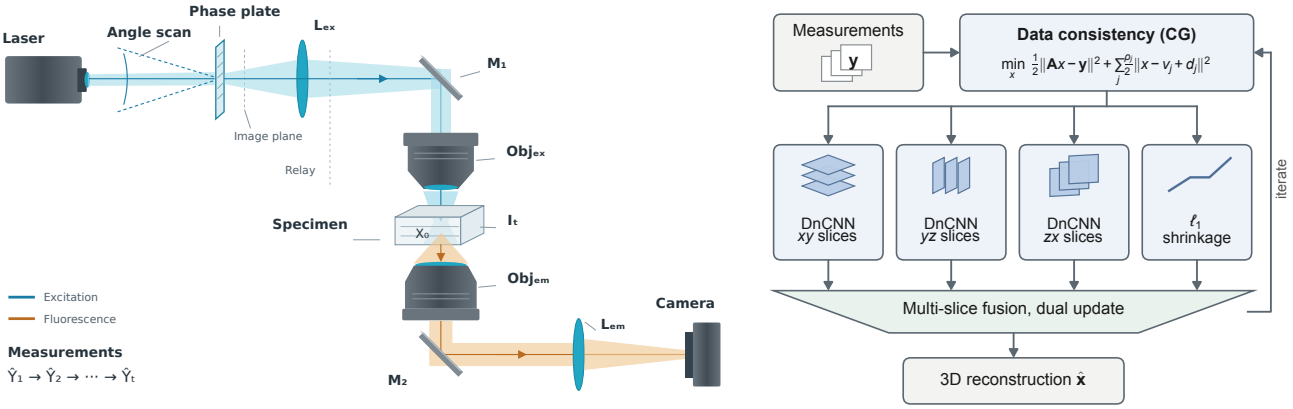


Figure 2. **Fixed-focus acquisition and reconstruction.** Left: a tilted plane wave illuminates a fixed phase plate placed 6.6 μm (specimen-space units) before an intermediate illumination image plane; an NA-0.8 vectorial relay maps the field into the specimen; a camera at a fixed detection focus records one image per illumination angle. Only the input angle changes: plate, specimen and focus are stationary, and nothing is scanned axially. Blue: excitation; orange: fluorescence. Right: one of the four decoders of Sec. 4, multi-slice plug-and-play recovery, alternates a conjugate-gradient data-consistency step with a 2D denoiser applied to the xy, yz and zx slices of a shared volume and an ℓ_1 shrinkage, fused by ADMM.

$(\mathbf{s}_t, \sin\theta_t)$, and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ for the simulated refractive index of one. Thus, the two coordinates control the orientation and signed magnitude of the transverse phase gradient. The scan contains 256 distinct directions with $0.5 \le \|\mathbf{s}_t\| \le \sqrt{3}/2$. Each incident field passes through the phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagates a distance D to the intermediate illumination image plane by angular-spectrum propagation. The NA-0.8 objective pupil filters the modulated field before a vectorial Richards–Wolf relay [19, 26] produces the specimen excitation $I_t = \sum_{c\in\{x,y,z\}} |E_{t,c}|^2$, assuming x-polarised illumination. Coordinates and distances are expressed in specimen-equivalent units. The camera records one fluorescence image per direction at a fixed detection focus. Acquisition is sequential, not single-shot.

Why the angle encodes depth. Propagation converts the plate's phase structure into spatially varying excitation. Scanning the incident angle shifts this pattern differently at different depths, giving each location a depth-dependent illumination sequence. An ideal thin-screen model illustrates the mechanism: without pupil clipping, a change Δs of the transverse direction cosine displaces the pattern at propagation distance d from the plate by $d\,\Delta s/\sqrt{1-\bar{s}^2}$ (paraxially $d\,\Delta s$), where $\bar{s}$ is the mean direction cosine and $d = D$ at the conjugate plane of focus. Different depths therefore translate at different rates, and plate defocus places the shear pivot outside the specimen. This relation provides intuition; the high-NA vectorial model includes the finite pupil and departures from pure translation, and Sec. 6.1 measures the actual shear.

Image formation. Let $X_0(\mathbf{r}, z)$ denote the specimen's fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z') X_0(\mathbf{r}', z') I_t(\mathbf{r}', z')\, \mathrm{d}^2\mathbf{r}'\, \mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity point-spread function and Ω is the specimen volume. Pixel p, occupying region D_p, records the normalised sample $\hat{Y}^0_{t,p} = |D_p|^{-1} \int_{D_p} g_t(\mathbf{r})\, \mathrm{d}^2\mathbf{r}$. Fluorescence contributions therefore add incoherently before pixel averaging. Multiplication by the excitation pattern mixes specimen spatial frequencies before detection blur, so the camera need not resolve the illumination pattern itself. Depth discrimination depends on how these measurements vary across illumination angles.

Voxel-integrated operator. To capture excitation variations within reconstruction voxels, we evaluate the optics on a finer grid and integrate the measurement responses over each voxel. Let B be the fine-grid measurement operator, including emission propagation and pixel averaging, and let U replicate each reconstruction coefficient over its fine-grid cells. Then $\mathsf{A} = \mathsf{B}\mathsf{U}$ preserves the fine-grid response for every specimen represented in this piecewise-constant basis.

Sub-voxel specimen structure remains unrepresented. For a fine-grid specimen $\mathbf{x}_f$, its coarse representation $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$, obtained by voxel averaging, satisfies

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{U}\mathsf{R}\mathbf{x}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{2}$$

Thus, integration retains within-voxel optical variation, while $\boldsymbol{\delta}_{\text{disc}}$ captures specimen approximation error.

Linear, incoherent fluorescence enables efficient assembly:

$$\mathsf{A}_{(t,p),j} = \sum_{i\in\mathcal{P}_j} I_t(\mathbf{r}_i, z_i)\, g_{p,i}, \tag{3}$$

where $\mathcal{P}_j$ contains voxel j's fine-grid cells, $(\mathbf{r}_i, z_i)$ is the location of cell i, and $g_{p,i}$ is its emission footprint at pixel p, including its integration weight. These footprints are computed once and reused across angles. After precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF\log F)$ for constructing columns through separate FFT-based forward simulations, with T patterns, P detector samples, F fine-grid cells and N reconstruction voxels.

Learning the plate. We optimise the plate phase ϕ to reduce correlations between measurement signatures at different depths, with the aim of improving depth discrimination. For a voxel at lateral position $\mathbf{r}$ and depth z, let $\mathbf{a}_{\mathbf{r},z}$ denote its centred, unit-norm operator column (Eq. (3)), collecting detector responses across illumination angles. We minimise

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r}} \sum_{z\neq z'} w_{zz'} \left(\mathbf{a}_{\mathbf{r},z}^{\top}\mathbf{a}_{\mathbf{r},z'}\right)^2, \tag{4}$$

where pairs of planes mirrored about the detection focus, $(z, 14-z)$ on the 16-plane grid, receive weight two because they are hardest to separate, and other pairs receive weight one. Squaring penalises correlations of either sign; normalisation prevents uniform amplification of the signatures from reducing the loss.

Training differentiates through the excitation, vectorial relay and detection model, without specimen examples or a reconstruction solver. The objective is a surrogate for depth discrimination, not a guarantee of stable recovery; whether it helps is tested with the decoders of Sec. 4, none of which is used in training. Training details are given in the supplement.

4. Decoders and depth intervals

The plate is trained without any decoder. We then decode its measurements with four reconstruction methods of different kinds. None of them is trained on measurements, so the same prior serves every encoder; only scalar weights are selected per encoder and compression, on validation volumes. We call them TV, pretrained PnP, learned PnP and diffusion.

TV. Box-constrained least squares with isotropic 3D total variation [27], solved by ADMM [5]: a non-learned regularity prior.

Multi-slice plug-and-play. We combine least-squares data fitting with multi-slice denoising and sparsity through plug-and-play ADMM [18, 30]. For orientation $o \in \{xy, yz, zx\}$, P_o rearranges the shared volume into slices, which a 2D denoiser $\mathcal{D}$ processes independently, giving the volume denoiser $\mathcal{D}_o = \mathsf{P}_o^{-1}\mathcal{D}\mathsf{P}_o$. With branches $\mathcal{J}$, each iteration solves

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathsf{A}\mathbf{x}-\mathbf{y}\|_2^2 + \sum_{j\in\mathcal{J}} \tfrac{\rho_j}{2}\|\mathbf{x}-\mathbf{v}_j^k+\mathbf{d}_j^k\|_2^2 \tag{5}$$

by conjugate gradients, followed by $\mathbf{v}_j^{k+1} = \mathcal{D}_j(\mathbf{x}^{k+1}+\mathbf{d}_j^k)$ for each denoising branch, soft thresholding for the sparsity branch, and dual updates $\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}$. *Pretrained PnP* uses three slice branches with a DnCNN [36] trained on natural images, plus the sparsity branch. *Learned PnP* replaces that DnCNN by a 2D network trained on neuron volumes in the three slice branches and adds a fourth branch with a 3D network trained on the same volumes [15, 37]; both are blind Gaussian denoisers. Iteration counts and the selection grid are fixed (supplement).

Diffusion. A 3D U-Net denoiser in the EDM parametrisation [16] is trained on the same neuron volumes. We sample with DiffPIR [38]: at each of 100 noise levels, the denoised estimate is pulled towards the data by a proximal step on the weighted least-squares term $\frac{1}{2}\|\mathsf{A}\mathbf{x}-\mathbf{y}\|_{\mathsf{W}}^2$, a Gaussian approximation of the Poisson likelihood with W the inverse photon-noise variance, solved by conjugate gradients. We report the mean of eight samples. We use the samples as a reconstruction method, not as calibrated posterior draws.

Conformal depth intervals. For each lateral column c of a reconstruction with signal, let $\hat{z}_c$ be its axial intensity centroid and z_c the reference one. Around each column we place the interval $\hat{z}_c \pm \lambda$, and define the loss $L(\lambda)$ of a volume as the signal-weighted fraction of its columns whose reference centroid falls outside. L is non-increasing in λ and bounded by one, so conformal risk control [2] applies: with mean loss $\hat{R}_n(\lambda)$ over n calibration volumes reconstructed exactly like the test volumes,

$$\hat{\lambda} = \inf\left\{\lambda : \tfrac{n}{n+1}\hat{R}_n(\lambda) + \tfrac{1}{n+1} \le \alpha\right\} \tag{6}$$

gives $\mathbb{E}[L(\hat{\lambda})] \le \alpha$ for a new volume, provided the calibration volumes and the new one are exchangeable. This controls the expected fraction of misplaced signal per volume; it is not a coverage statement for every column, for every specimen, or for connectivity.

Bounds without the learned prior. The data together with intensity bounds define a measurement-consistency set $\mathcal{C}(\mathbf{y}) = \{\mathbf{x}\in[0,1]^N : \ell_i \le (\mathsf{A}\mathbf{x})_i \le h_i\}$, where the per-row intervals contain the noise-free measurement with stated joint probability under the noise model. Per-voxel bounds $\min_{\mathcal{C}} x_v$ and $\max_{\mathcal{C}} x_v$ (two linear programs per voxel) show how far any consistent volume can move. As intuition for

why the plate objective should narrow them, consider two voxels with unit-norm columns of correlation ρ in an ℓ_2 ball of radius ε: the range of either voxel is $\pm\varepsilon/\sqrt{1-\rho^2}$, from the Schur complement of the 2×2 Gram matrix. This simplified model ignores the coupling of many voxels, the box and the difference between the centred columns of the loss and the raw columns of A; we measure the bounds directly on a reduced field where exact optimisation is feasible.

5. Experimental setup

All experiments are simulations with the optics of Sec. 3: 532 nm excitation and detection, excitation NA 0.8, detection NA 0.1, a phase plate with $0.32\,\mu$m features at $D = 6.6\,\mu$m, 256 directions, $0.08\,\mu$m optical and $0.32\,\mu$m analysis grids, $0.64\,\mu$m reconstruction voxels, and linear fluorescence in a homogeneous non-scattering medium.

Volumes and compression. The reconstruction grid is $16 \times 48 \times 48$ ($N = 36{,}864$, $30.7 \times 30.7 \times 10.2\,\mu\text{m}^3$) with 12×12 camera samples per image, so $M = 144k$ for k directions and the compression is $C = N/M \in \{1, 4, 8\}$ for $k = 256, 64, 32$ (five random direction subsets for $C > 1$). Reference volumes are two-photon stacks of mouse pyramidal neurons from the public DEEP^2 data [31], min–max normalised and resampled to the analysis grid. The test set holds two prespecified crops from each of six cells of three animals, chosen by a structure score; for the decoder comparison (Sec. 6.2) we added eight crops per cell drawn at random from the eligible ones, after all weights had been selected (60 crops). Other analyses use the 12 prespecified crops. The neuron-trained denoisers and the diffusion model are trained on 459 crops of a fourth animal; crops of a held-out stack of that animal, simulated like the test crops, calibrate the conformal intervals (35 non-overlapping crops; all 240 overlapping crops as a sensitivity check). Scalar weights of every decoder are selected per encoder and compression on two vascular crops of a separate stack, on grids that extend automatically at their edges. No test volume is used for training, tuning or calibration.

Encoders and photon budget. The random plate (three realisations), a defocused binary amplitude mask at the plate's position, the same scan without a plate, and an in-focus DMD with 256 random binary patterns share relay, pupil, detector and assembly. Their excitation is simulated on an optical grid larger than the field, so beam edges lie outside it, and normalised to the same deposited dose. Counts are $\text{Poisson}(\kappa_0\, 256\, y_i/k)$: a fixed total budget shared by the k frames, with κ_0 set per specimen so that the random plate reaches 15 dB at $k = 256$ (1000 photons per pixel per frame at the mean signal). The learned plate starts from random realisation 0 and changes only inside the specimen field; it is trained with two seeds. On one RTX 4500 Ada GPU, 1000 training steps take 20 min, assembling the plate's operator 6 min, and one reconstruction at $C = 8$ 4–6 s (PnP) or 17 s (TV).

Metrics and statistics. Besides PSNR we report foreground SSIM and the *local depth error*: the mean absolute difference between the true and reconstructed axial centroids of each lateral column, weighted by the true column brightness over columns above 10 % of the brightest. Two references calibrate it: a depth-blind oracle with the exact lateral structure and each column spread over the crop's mean axial profile scores $1.32\,\mu$m, and a flat estimate, uniform in depth, $2.15\,\mu$m. Crops of one cell are not independent, so intervals resample cells (six cells, two crops each, 10^4 resamples) and encoders are paired on the same objects, direction subsets, dose and noise protocol; decoders of one encoder share its measurements.

Reduced field. Exact bounds over $\mathcal{C}(\mathbf{y})$ use dense linear programs, so they run on a reduced field of the same optics ($4 \times 12 \times 12$ voxels, 3×3 pixels, $k = 32$) with Bonferroni intervals at $\alpha = 0.05$ and Gaussian noise at 15 dB, on 24 synthetic phantoms with three noise seeds and on 48 windows cut from the test neurons (supplement).

6. Results

6.1. Defocus under the angle scan encodes depth

We first test the untrained mechanism. Figure 3 compares the random plate with three encoders that share its relay, detector, dose, volumes, decoders and tuning protocol. A robust line fit of the cross-correlation shift per depth gives shear slopes of 0.74–0.94 of the thin-screen prediction, with the pivot near the plate's conjugate plane. Per-pixel excitation sequences decorrelate within about $1\,\mu$m of depth, and point emitters are assigned a wrong depth in 1.5–1.8 % of cases (three plates) against 4.4–5.3 % (three defocused masks), 25 % (DMD, half at the mirror depth) and 89 % without the plate. The in-focus DMD stays at the depth error of a flat estimate (2.17 against $2.15\,\mu$m) while the plate reaches $0.36\,\mu$m (pretrained PnP, $C = 1$). The defocused amplitude mask comes close to the plate: over three realisations of each, the plate leads in PSNR in all nine pairings ($+0.4$ to $+1.2$ dB at $C = 8$) but not in depth error. Depth is therefore encoded by defocus under the angle scan, and the random plate is already a strong baseline; the question is how much learning adds.

6.2. The learned plate under four decoders

Figure 1 and Tab. 1 decode the learned plate's measurements with each method of Sec. 4. On the 12 prespecified crops, the learned plate lowers the depth error at $C = 8$ by 37 %

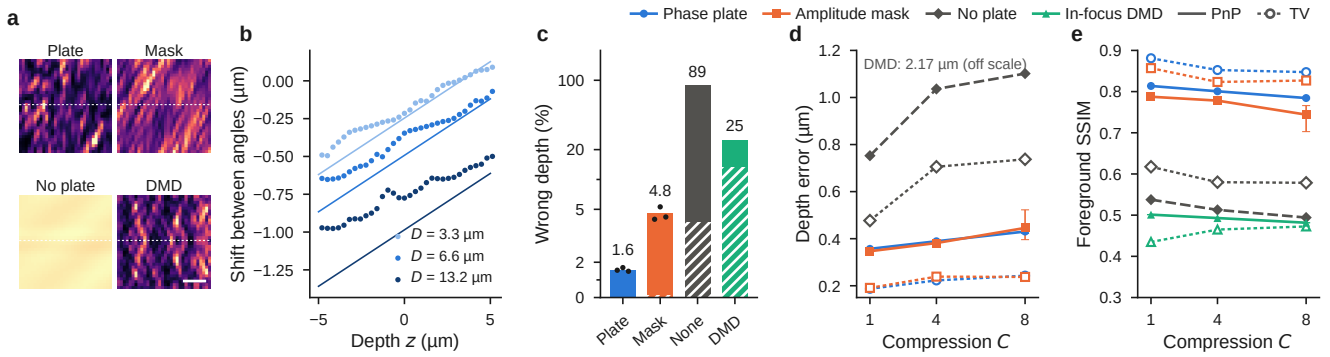


Figure 3. **Defocus under the angle scan encodes depth; in-focus patterns and a plain scan do not** (full-size field, equal deposited excitation dose, overfilled illumination aperture; phase plate: the random plate). (a) One excitation pattern per encoder, xz section through the specimen depth ($\pm 5\,\mu$m, focus dotted), each tile scaled to its own 99.5th percentile. (b) Lateral shift of the plate's excitation between two scan angles at every depth, from the cross-correlation peak, for plate defocus $D = 3.3$, 6.6 and 13.2 μm; lines: thin-screen prediction of Sec. 3. (c) Point emitters assigned a wrong depth by maximum likelihood with unknown brightness at $k = 32$ (dots: three plate and three mask realisations; hatched: mirror depth). (d, e) Reconstructions of 12 neuron crops from 6 cells: local depth error and foreground SSIM versus compression, pretrained PnP (solid, no plate dashed; plate and mask at $C = 4, 8$: mean and range over three realisations) and TV (dashed); the DMD's depth error of 2.17 μm is off scale. No plate: simulated aperture four times the field in (a, d, e), twice in (c). All values are tabulated in the supplement.

	random plate		learned plate		depth change vs random		DMD
decoder	PSNR	depth	PSNR	depth	seed 0 [95 % CI]	seed 1	depth†
TV	36.70	0.259	38.29	0.150	−42 % [−46, −38]	−43 %	2.31
pretrained PnP	35.72	0.490	37.56	0.224	−54 % [−57, −51]	−55 %	2.17
learned PnP	38.01	0.197	39.60	0.139	−30 % [−32, −27]	−31 %	2.25
diffusion*	34.34	0.455	38.74	0.168	−63 % [−70, −54]	−64 %	2.37

Table 1. **The learned plate under four decoders** ($C = 8$, mean over 60 test crops from six cells × 5 measurement draws, equal dose; PSNR in dB, local depth error in μm). Plates are paired on the same objects, direction subsets, dose and noise protocol; depth change with cell-bootstrap 95 % interval. *Descriptive, 12 prespecified crops (Sec. 6.2). †DMD on the 12 prespecified crops (flat estimate 2.15 μm).

with TV, 47 % with pretrained PnP and 26 % with learned PnP (supplement). Because these crops were chosen by a structure score, we then drew 48 more crops at random from the same six cells and reconstructed them with the weights already selected. On these new crops the reductions are 43, 56 and 31 %, and depth improves on every crop under every decoder. Over all 60 crops the reductions are 42, 54 and 30 % with intervals that exclude zero, PSNR rises by 1.6–1.8 dB, and a second training seed gives 31–55 %. On the prespecified crops the gain holds at every compression we tested: −25 to −37 % with TV, −45 to −48 % with pretrained PnP and −26 to −32 % with learned PnP at $C =$ 1, 4 and 8 (supplement). Diffusion, run on the prespecified crops, shows the same direction (0.455 → 0.168 μm), but on the random plate it missed the criterion we fixed in advance for a competitive decoder (within 10 % of learned PnP's depth error), which inflates its relative gain; we report it as descriptive.

Fewer frames. On the prespecified crops and at the same total dose, the learned plate with 32 frames is better than the random plate with all 256 frames under pretrained PnP (−35 % [−42, −28]) and has a lower mean under TV (0.153 against 0.187 μm). Under learned PnP the means are similar (0.164 against 0.161 μm), but the interval [−17, +15] % is too wide to establish equivalence.

Closer performance across decoders. Over the 60 crops, the mean depth errors of TV and the two PnP decoders span 0.20–0.49 μm with the random plate and 0.14–0.22 μm with the learned plate. This is closer average performance, not a measure of agreement between the reconstructed volumes, which we have not tested.

6.3. Choice of plate objective

Table 2 trains the plate with three alternative decoder-free objectives under the same start, sampling and 1000 steps: a smooth worst-case coherence of the depth columns, following Miniscope3D [33]; a Poisson Fisher-information objective on depth with unknown brightness, following [28]; and our loss without centring. All improve on the random plate. Coherence and Fisher information keep 50–63 % of our depth gain over it: paired on the same measurements, our objective lowers their depth error by 12 and 14 % at $C = 8$ (intervals exclude zero; 11 and 10 of 12 crops) and by 19 and 14 % at $C = 1$. Centring made no difference here.

	$C=1$		$C=8$		ours vs this, $C=8$
plate objective	PSNR	depth	PSNR	depth	depth [95 % CI]
none (random plate)	39.18	0.161	37.11	0.221	−26 % [−35, −18]
worst-case coherence	39.95	0.135	38.20	0.186	−12 % [−16, −9]
depth Fisher information	40.15	0.128	38.15	0.190	−14 % [−19, −10]
depth decorrelation, uncentred	40.86	0.108	38.85	0.164	0 % [−1, +1]
depth decorrelation (ours)	40.82	0.109	38.83	0.164	–

Table 2. **Plate objectives** with the same start, sampling, step budget and seed, decoded with learned PnP (12 test crops; depth in μm). Each plate has its own validation tuning. Last column: depth change of our plate against each row, paired by crop, with cell-bootstrap interval.

	conformal depth interval ($\alpha=0.1$, $C=8$)			bound width
plate	$\pm\hat{\lambda}$ (μm)	vs random [95 % CI]	test miscov.	reduced field
random	0.646	–	0.060	1
learned, seed 0	0.474	−26.7 % [−30.1, −22.9]	0.075	0.79
learned, seed 1	0.486	−24.8 % [−29.6, −21.6]	0.060	0.79

Table 3. **Depth intervals and bounds.** Conformal half-width of learned-PnP reconstructions, calibrated on 35 non-overlapping held-out crops (calibration-bootstrap 95 % interval), and the realised miscoverage on the 12 test crops × 5 draws (signal-weighted, target $\alpha=0.1$). Bound width: mean per-voxel width over $\mathcal{C}(\mathbf{y})$ on the reduced field (15 dB, 72 queries), relative to the random plate.

This is a controlled comparison of our implementations of these objectives (one seed, learned PnP, two compressions), not a ranking of the objectives in general or of the imaging systems they come from; none was tuned beyond the shared budget.

6.4. Calibrated depth intervals

Conformal intervals. With $\alpha=0.1$, the half-width $\hat{\lambda}$ of Eq. (6) is $0.47\,\mu$m with the learned plate and $0.65\,\mu$m with the random plate, 27 % narrower (second seed: 24.8 %; all 240 overlapping crops: 26 %). The theorem guarantees that the expected signal-weighted fraction of misplaced columns is at most 10 % for volumes exchangeable with the calibration set. Our calibration crops come from one held-out stack of the animal used to train the priors, and the test cells come from three other animals, so exchangeability is an assumption rather than a design property. Empirically the intervals held on the test cells (realised miscoverage 0.06–0.075), which supports, but does not prove, that they transfer across animals.

Bounds without the learned prior. On the reduced field, using only the measurements, the intensity box and the noise model, the per-voxel bounds over $\mathcal{C}(\mathbf{y})$ are 21 % narrower with the learned plate, close to the $0.80\times$ suggested by the two-column intuition of Sec. 4 for the operator's drop in mean depth correlation from 0.868 to 0.761. Over 12.5–25 dB the narrowing equals 1.9 dB of SNR. On 24 sparse synthetic phantoms with three noise draws each, connections certified by the measurements alone rise from 3 to 12 of 48 connected instances at 15 dB; the gain comes from four phantoms and none is lost, but the noise draws of one phantom are not independent, so we do not attach a per-instance significance. On 48 windows cut from the test neurons, no voxel is determined by any plate at 15–45 dB: at eight-fold compression dense neuronal tissue is not determined by the measurements alone. Statements about neuronal structure therefore need a calibrated prior, and we limit ours to depth placement.

6.5. An in-focus encoder leaves decoders without depth

With the in-focus DMD every decoder stays at 2.17–2.37 μm, the error of a flat estimate (Fig. 1 and Tab. 1); on the 48 random crops learned PnP reaches $1.81\,\mu$m, above the depth-blind oracle (1.42) and near the flat estimate ($1.93\,\mu$m) of those crops. A second, independent DMD pattern set, run with TV and both PnP decoders, reproduces them to within 0.06 dB and $0.07\,\mu$m. Splitting each column's error into its mean over depth (lateral error) and the zero-mean remainder (depth error) shows where the prior's gain goes. At $C=8$ learned PnP shrinks the random plate's lateral error to $0.48\times$ of pretrained PnP's and its depth error to $0.71\times$ (median over crops; lower on 12 of 12), but for the DMD it shrinks the lateral error to $0.77\times$ and leaves the depth error at $1.04\times$, which is 99 % of the DMD's total. Figure 4 shows the consequence: the DMD's dendrites look sharper in xy, and its xz section remains without depth. A prior could in principle guess depth from population statistics, but these measurements give it nothing to confirm the guess, and none of the tested priors recovered it.

6.6. Calibration and tolerance

A learned plate is useful only if its gain survives fabrication and alignment. With the operator rebuilt from the fabricated plate, rounding the phase to four levels or adding 0.5 rad RMS height error to an 8-level plate keeps a depth reduction of 25.5 and 25.2 %, against 25.9 % for the nominal plate (learned PnP, $C=8$). A plate $0.5\,\mu$m closer or farther than designed keeps 27.8 and 26.4 % after recalibration, and scan-pointing errors up to 0.01 in direction cosine keep 18–22 % without recalibration. Uncalibrated plate-distance and lateral plate errors shift the excitation code almost rigidly (correlation ≥ 0.97 after the offset); they cost both plates 0.15–0.41 μm and remove the learned plate's advantage, so they must be registered, *e.g.* with one bead. Simulating the optics at the reconstruction spacing instead of integrating over voxels raises the forward discrepancy from 1.07 to 25.7 % (supplement).

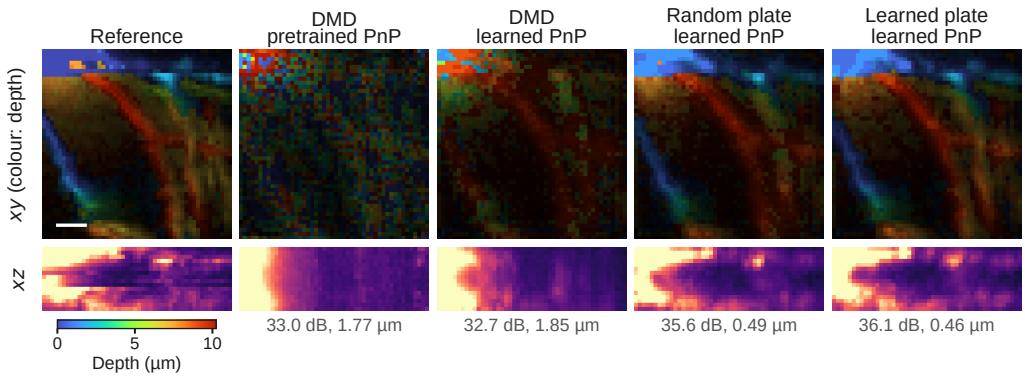


Figure 4. **Same decoder, different encoders** ($C = 8$, prespecified display crop with the largest reference depth spread, draw 0, identical display). Top: lateral maximum-intensity projection, colour coding the depth of the brightest voxel; bottom: xz projection; below: PSNR and local depth error of this crop. Learned PnP sharpens the DMD's lateral structure but not its depth. On this crop the learned plate improves on the random plate only slightly (0.46 against 0.49 μm); the cohort mean is in Tab. 1. Learned-plate column: reconstruction from the conformal study's rerun, which reproduces the recorded row to 0.02 dB. Bar: 5 μm.

7. Discussion and limitations

Training without a decoder. The plate is fabricated once, while decoders are replaced as priors improve. A plate optimised end-to-end with one network is matched to that network, and its benefit to another decoder must be tested anew. Our plate was trained without a decoder and improved depth under a regulariser, two plug-and-play decoders and, descriptively, a diffusion decoder. Four methods, two of which share an algorithm and two of which share training data, do not establish transfer to every future decoder; they show that the gain is not tied to one. We did not compare with end-to-end design, which asks for the best pair of optics and decoder rather than optics that serve several; our comparison is with decoder-free objectives under the same budget (Tab. 2).

What the gain is and is not. The learned plate improves depth placement, and conformal calibration turns the improvement into narrower intervals with an expected-risk guarantee under exchangeability. It does not make neuronal topology identifiable: at eight-fold compression no plate determines dense neuronal windows from the measurements alone (Sec. 6.4). In supplementary experiments, conformal connectivity calls on these reconstructions missed their nominal error rate, the diffusion samples were over-confident (per-voxel two-standard-deviation coverage 0.71–0.75) and did not reduce confident wrong connectivity calls, and query-adaptive direction selection did not meet its prespecified criterion. Connections drawn from such reconstructions remain the prior's choice.

Limitations. All results are simulations, without hardware validation, of linear fluorescence in a homogeneous non-scattering medium with an ideal relay. A uniform background with read noise, or one tenth of the dose, keeps a 27–55 % gain on every prespecified crop (supplement), but scattering and aberrations will reduce every encoder's depth information. The test set has 60 crops from six cells of three animals, which are not independent; bootstrap intervals over six cells are approximate, although the learned plate lowers the depth error on 59–60 of 60 crops under TV and both PnP decoders. Diffusion, the conformal study and the objective comparison use the 12 prespecified crops. The plate is trained with two seeds and the priors on one animal. The conformal calibration uses one held-out stack of that animal; a different tissue or preparation would need its own calibration. Plate distance and lateral position must be registered before use.

8. Conclusion

A phase plate scanned in angle before an intermediate image plane gives each depth its own excitation sequence, and a specimen-free objective on the resulting measurement signatures improves that encoding without a decoder in the loop. In simulation, the learned plate lowered the depth error of TV and two plug-and-play decoders by 30–54 %, pointed the same way for a diffusion decoder, and narrowed calibrated depth intervals by 27 %, while an in-focus encoder left every tested decoder without depth. Learned optics can be trained for what the measurements record and then tested across decoders, rather than being matched to one.

LLM Usage Statement

LLMs were used for grammar and fact checking.

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